

Coding & Solution

Date	01 July 2026
Team ID	SWTID-2026-3278
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/yuvapriya-1/Credit-Card-Approval-Predictor
Programming Language(s)	Python
Framework(s) Used	Scikit-learn, Pandas, NumPy, Flask
Key Features Implemented	Data preprocessing, Missing value handling, Feature encoding, Model training, Credit card approval prediction, Performance evaluation

Pending / Incomplete Features	• User login and authentication • Real-time bank database integration • Email/SMS notification for approval status • Cloud deployment
Setup / Run Instructions	☐ Install Python 3.10 or later. ☐ Install required libraries: <code>pip install pandas numpy scikit-learn flask</code> ☐ Run the application: <code>python app.py</code> ☐ Upload applicant details or enter data. ☐ Click Analyze to view the credit card approval



Code Quality Checklist

S.N o	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The machine learning model was trained using a supervised classification approach and achieves a high accuracy rate in identifying creditworthiness. The model is optimized to evaluate risk factors such as income, employment status, and credit history. Future enhancements will focus on integrating real-time API data and advanced hyperparameter tuning for further precision.